2.3.2 Fiber Optic Facts

To connect computers using fiber optic cables, you need two fiber strands. One strand transmits signals, and the other strand receives signals. Long-haul runs sometimes only need one fiber. The send and receive signals are transmitted over the same fiber. The following are the components of fiber optic cabling:

- The core carries the signal. It is made of glass or plastic.
- The cladding maintains the signal in the center of the core as the cable bends.
- The sheathing protects the cladding and the core.

Fiber optic cabling offers the following advantages and disadvantages:

Advantages	Disadvantages
▪ Totally immune to EMI (electromagnetic interference) ▪ Highly resistant to eavesdropping ▪ Supports extremely high data transmission rates ▪ Allows greater cable distances without a repeater	▪ Very expensive ▪ Difficult to work with ▪ Special training required to attach connectors to cables

Multi-Mode vs. Single-Mode

Multi-mode and single-mode fiber cables are not interchangeable. The following table describes multi-mode and single-mode fiber cables.

Type	Description
Single-Mode	▪ Data transfers through the core using a single light path. ▪ The core diameter is around 8–10.5 microns. ▪ Cable lengths can extend a great distance. ▪ There is less modal dispersion, so bandwidths can be higher. ▪ Higher-cost electronics are required to send signals down a single path. ▪ Optimized for 1310 nm and 1550 nm light sources.
Multi-Mode	▪ Data transfers through the core using multiple light paths. ▪ The core diameter is around 50 to 100 microns. ▪ There is more modal dispersion due to the multiple paths. ▪ Cable lengths are limited in distance and are dependent on bandwidth. ▪ Higher light gathering capacity simplifies connections and allows lower-cost electronics. ▪ Optimized for 850 nm and 1300 nm light sources.

Connector Types

Fiber optic cabling uses the following connector types:

Type	Description
ST Connector	▪ Used with single-mode and multi-mode

	cabling. ▪ Has a keyed bayonet-type connector. ▪ Also called a push-in and twist connector. ▪ Each wire has a separate connector. ▪ Nickel plated with a ceramic ferrule to ensure proper core alignment and prevent light ray deflection. ▪ Mnemonics are Set-and-Twist or Straight Tip.
SC Connector 	▪ Used with single-mode and multi-mode cabling. ▪ Has a push-on/pull-off connector that uses a locking tab to maintain connection. ▪ Each wire has a separate connector. ▪ Uses a ceramic ferrule to ensure proper core alignment and prevent light ray deflection. ▪ Mnemonics are Set-and-Click or Square Connector.
LC Connector 	▪ Used with single-mode and multi-mode cabling. ▪ Composed of a plastic connector with a locking tab that is similar to a RJ45 connector. ▪ A single connector with two ends keeps the two cables in place. ▪ Uses a ceramic ferrule to ensure proper core alignment and to prevent light ray deflection. ▪ Is half the size of other fiber optic connectors. ▪ Mnemonics are Lift-and-Click or Little Connector.
MTRJ Connector 	▪ Used with single-mode and multi-mode cabling. ▪ Composed of a plastic connector with a locking tab. ▪ Uses metal guide pins to ensure that it is properly aligned. ▪ A single connector with one end holds both cables. ▪ Uses a ceramic ferrule to ensure proper core alignment and prevent light ray deflection.
FC Connector	▪ Typically used with single-mode cabling. ▪ Each wire has a separate connector. ▪ Uses a threaded connector.

	▪ Designed to stay securely connected in environments where it may experience physical shock or intense vibration.

For long cable running between floors or overhead, you might hire an experienced contractor to install the cable and the necessary connectors. Adding connectors onto a fiber optic cable takes some practice. Remember to complete the following:

- Keep the area as clean as possible.
- Cut the cable with a clean 90-degree cut.
- Polish the end of the cable prior to adding the connector. Use special polishing film and tools for polishing cable ends.
- Glue or crimp the connector onto the cable.
- Cover or cap any connectors that won't be hooked up immediately.
- If necessary, you can directly splice two cable ends together; however, this requires expensive and specialized equipment.

Fiber Optic Cable Facts

There are two light source technologies prevalent in fiber optic communications, diode laser and high-radiance light-emitting diode (LED). The light produced by these technologies is in the infrared region of the light spectrum.

- The most common wavelengths used in fiber optics are 850 nm, 1300 nm, 1310 nm and 1550 nm.
  - In glass, these longer wavelengths have lower attenuation or signal loss due to scattering.
  - Attenuation in glass due to light absorption for these wavelength is almost zero.
- Multi-mode fiber is designed to operate at 850 nm and 1300 nm.
- Single-mode fiber is optimized for 1310 nm and 1550 nm.

Wave Division Multiplexing (WDM) joins several light wavelengths (colors) onto a single strand of fiber.

- This enables light signals in both directions across a single fiber.
- Today's systems can easily multiplex 160 signals.
- WDM is mostly used by long-haul and high-speed providers.
- Most WDM systems are designed to be used with single-mode fiber.

When working with fiber optic cabling, you can use media converters to switch between different network media. For example, you can convert:

- Single-mode fiber to copper Ethernet wiring
- Multi-mode fiber to copper Ethernet wiring
- Single-mode or multi-mode fiber to coaxial wiring
- Single-mode fiber to multi-mode fiber

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